

## SAVEETHA ENGINEERING COLLEGE

*(Autonomous Institution — Affiliated to Anna University)*

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SUBJECT

**Prompt Engineering**

## EXPERIMENT NO. 7

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**Reproducing an Image Using Prompts for Image Generation**

### STUDENT DETAILS

|                            |                              |
|----------------------------|------------------------------|
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| <b>Subject</b>             | Prompt Engineering           |
| <b>Institution</b>         | Saveetha Engineering College |

## EXPERIMENT NO. 7

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### Reproducing an Image Using Prompts for Image Generation

#### Aim

To demonstrate the ability of text-to-image generation tools to reproduce an existing image by crafting precise prompts.

#### Objective

To identify key visual elements within a reference image and use prompt engineering techniques to generate an image that closely resembles the original using AI image generation tools.

#### AI Tools Used

- ChatGPT Image Generation
- Google Gemini
- Bing Image Creator
- Leonardo AI
- Canva AI
- Midjourney

#### Reference Image 1: Butterfly Coloring Page

##### Visual Elements Identified

##### i) Objects / Subject

- Large butterfly positioned at the center
- Symmetrical wings with decorative patterns
- Coloring-book style outline

##### ii) Background

- Plain white background
- No additional objects or scenery

##### iii) Lighting

- Flat lighting
- No shadows or highlights

##### iv) Style

- Black-and-white coloring page
- Clean vector line art
- Children's coloring book design

#### v) Color Combination

- Monochrome (black outlines on white background)

#### vi) Image Quality

- High-resolution vector illustration
- Sharp edges and clear outlines

### Initial Prompt

**Prompt:** Create a simple butterfly coloring page with black outlines on a white background.

### Refined Prompt

**Prompt:** Create a high-resolution butterfly coloring page featuring a large symmetrical butterfly centered on a plain white background. Use clean black vector outlines with detailed decorative patterns inside the wings. The illustration should resemble a professional children's coloring book page. No colors, shading, shadows, or background elements. Minimalist black-and-white line art, crisp edges, printable quality, 4K resolution.

### Expected Output

A butterfly coloring page closely matching the reference image with:

- Symmetrical wings
- Detailed wing patterns
- White background
- Clean black outlines
- Printable coloring-book appearance

## Reference Image 2: Product Launch Poster

### Visual Elements Identified

#### i) Objects / Subject

- Modern product launch advertisement
- Product bottle placed prominently
- Promotional text and branding elements

#### ii) Background

- Gradient background
- Abstract geometric shapes
- Professional marketing design

#### iii) Lighting

- Studio lighting
- Product spotlight effect
- High contrast

**iv) Style**

- Commercial advertising poster
- Corporate branding
- Modern graphic design

**v) Color Combination**

- Blue and orange gradients
- White typography
- Bright accent colors

**vi) Image Quality**

- High-definition poster design
- Professional marketing template
- Sharp typography and graphics

## Initial Prompt

**Prompt:** Create a product launch poster with a product image and promotional text.

## Refined Prompt

**Prompt:** Design a professional product launch poster featuring a premium cosmetic bottle placed at the center. Use a modern blue and orange gradient background with abstract geometric elements. Add clean white typography, bold headline reading 'NEW PRODUCT LAUNCH', professional branding layout, studio-quality lighting, spotlight effect on the product, high contrast, marketing campaign style, corporate advertisement design, ultra-high-resolution poster, realistic shadows, premium commercial look.

## Algorithm

1. Select the Reference Image - Choose or use the provided image for reproduction.
2. Analyze the Image - Identify major components and visual details.
3. Create Initial Prompt - Write a basic text-to-image prompt.
4. Generate Initial Output - Use an AI image generation tool to generate the image.
5. Refine the Prompt - Improve the prompt using detailed descriptions:
  - Add style
  - Add lighting
  - Add texture
  - Add composition details

- 6. Generate Final Output - Produce an improved image output using the refined prompt.
- 7. Compare the Results - Compare the original and generated images and document similarities, differences, and improvements achieved through prompt refinement.

Comparison Table

| Feature               | Original Image | Initial Output | Final Output |
|-----------------------|----------------|----------------|--------------|
| Subject Accuracy      | High           | Medium         | High         |
| Background Similarity | High           | Medium         | High         |
| Style Match           | High           | Low            | High         |
| Lighting Accuracy     | High           | Medium         | High         |
| Color Accuracy        | High           | Medium         | High         |
| Overall Similarity    | 100%           | 60-70%         | 90-95%       |

Observation

The initial prompts generated images with basic similarity to the reference images. After refining the prompts by including detailed information about composition, lighting, style, color schemes, and image quality, the generated outputs became significantly closer to the original images.

Result

Thus, the original images were successfully reproduced using prompt engineering techniques and AI image generation tools. The experiment demonstrated that detailed and well-structured prompts improve the accuracy and quality of AI-generated images, resulting in outputs that closely resemble the reference images.